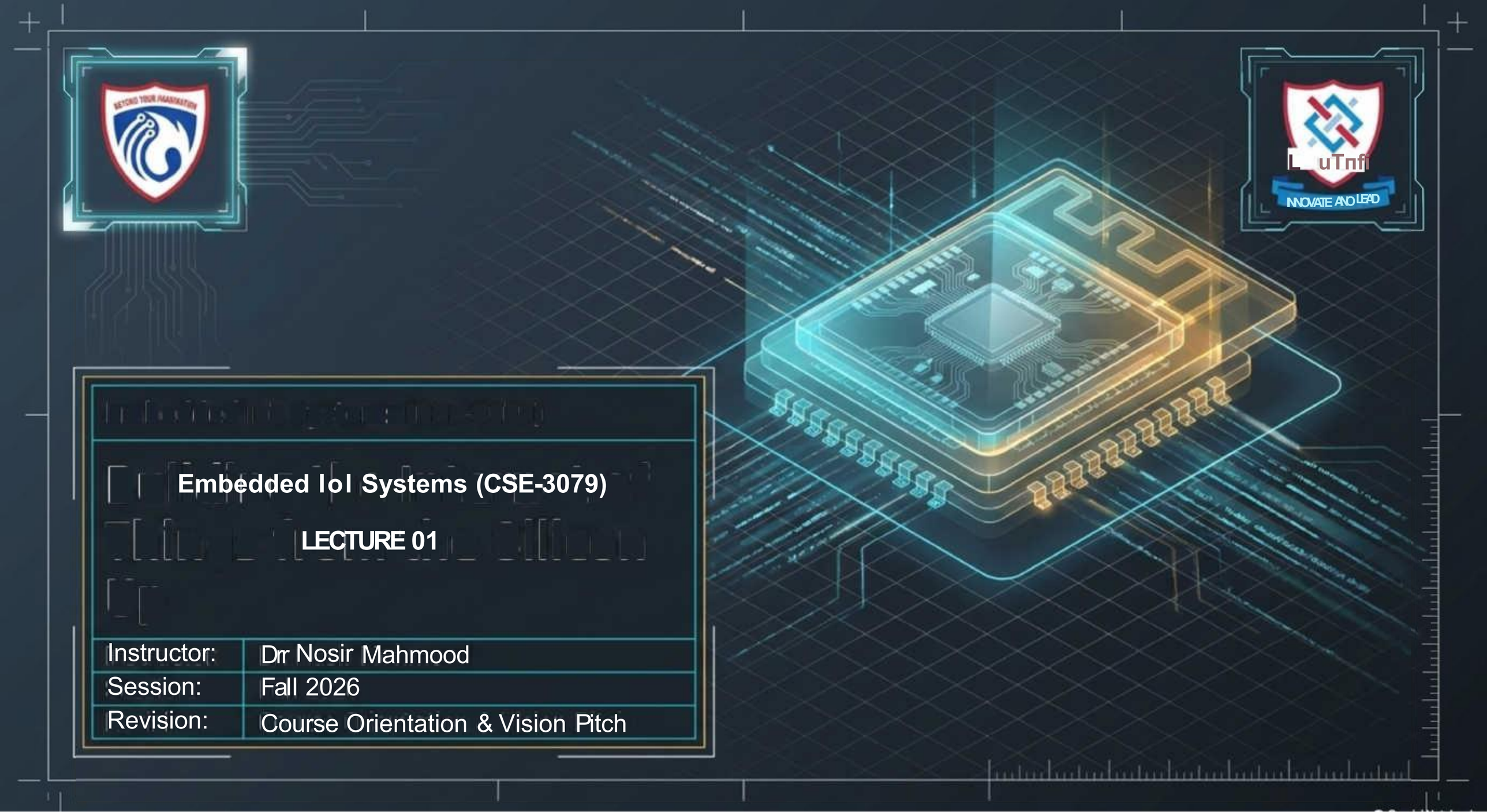
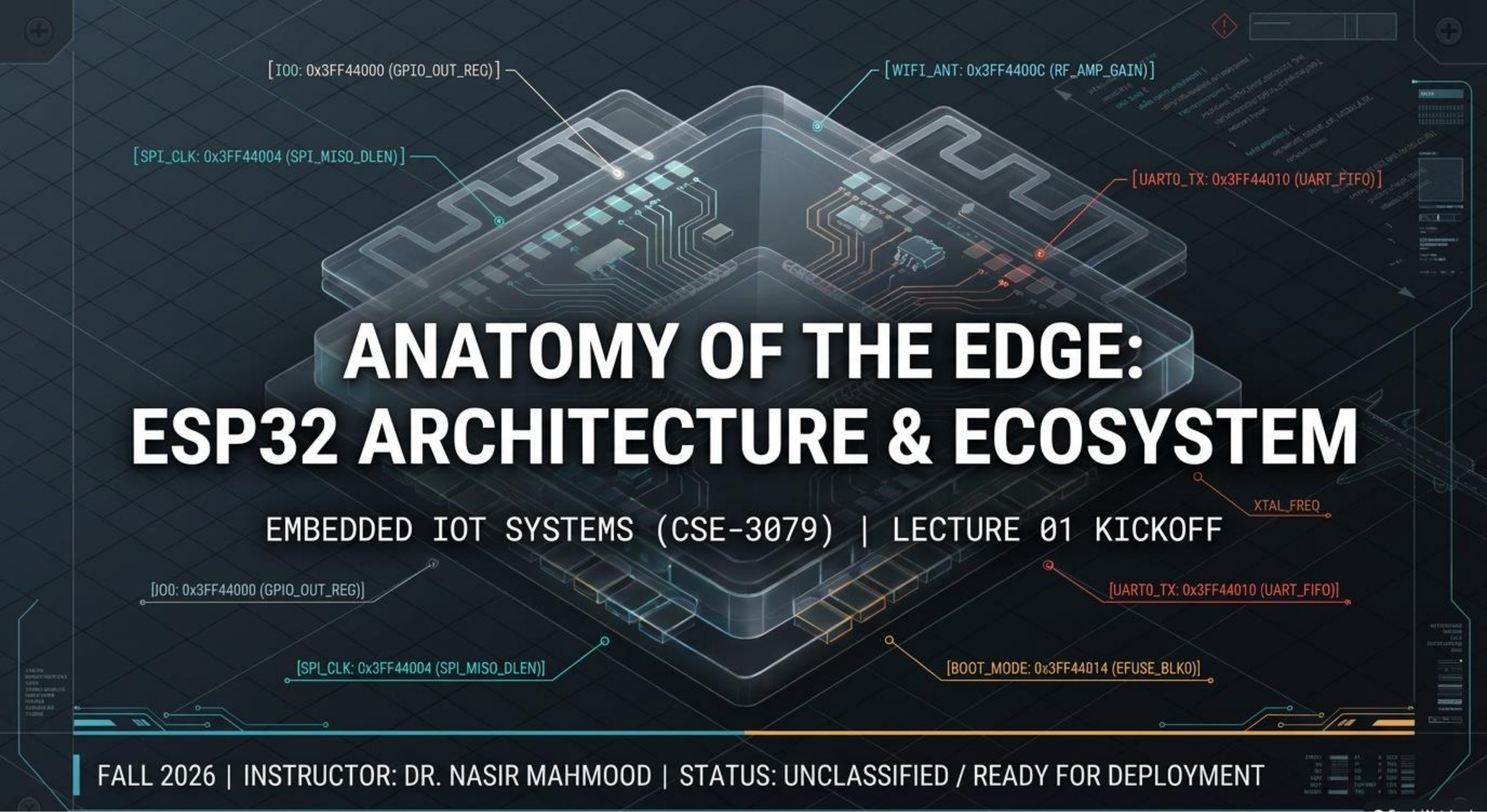


Embedded Iol Systems (CSE-3079)

LECTURE 01

Instructor:	Drr Nosir Mahmood
Session:	Fall 2026
Revision:	Course Orientation & Vision Pitch



ANATOMY OF THE EDGE: ESP32 ARCHITECTURE & ECOSYSTEM

EMBEDDED IOT SYSTEMS (CSE-3079) | LECTURE 01 KICKOFF

FALL 2026 | INSTRUCTOR: DR. NASIR MAHMOOD | STATUS: UNCLASSIFIED / READY FOR DEPLOYMENT

Software is Blind Without Hardware: The Convergence Catalyst

Physical Reality

Heat

Motion

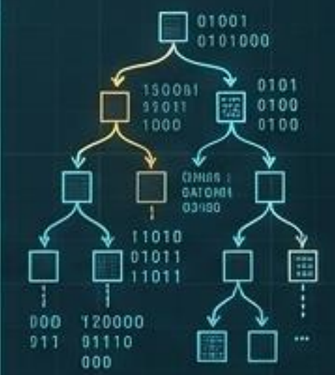
Light

Pressure

Edge Silicon



Digital Intelligence



Algorithmic



Data stream

The Edge Node

Translating physical analog signals into digital data. Requires low power, high connectivity, and robust processing.

The Career Leap

The hardware-software convergence requires engineers who can orchestrate both ends of the spectrum—from smart hardware manipulation to cloud-scale edge computing.

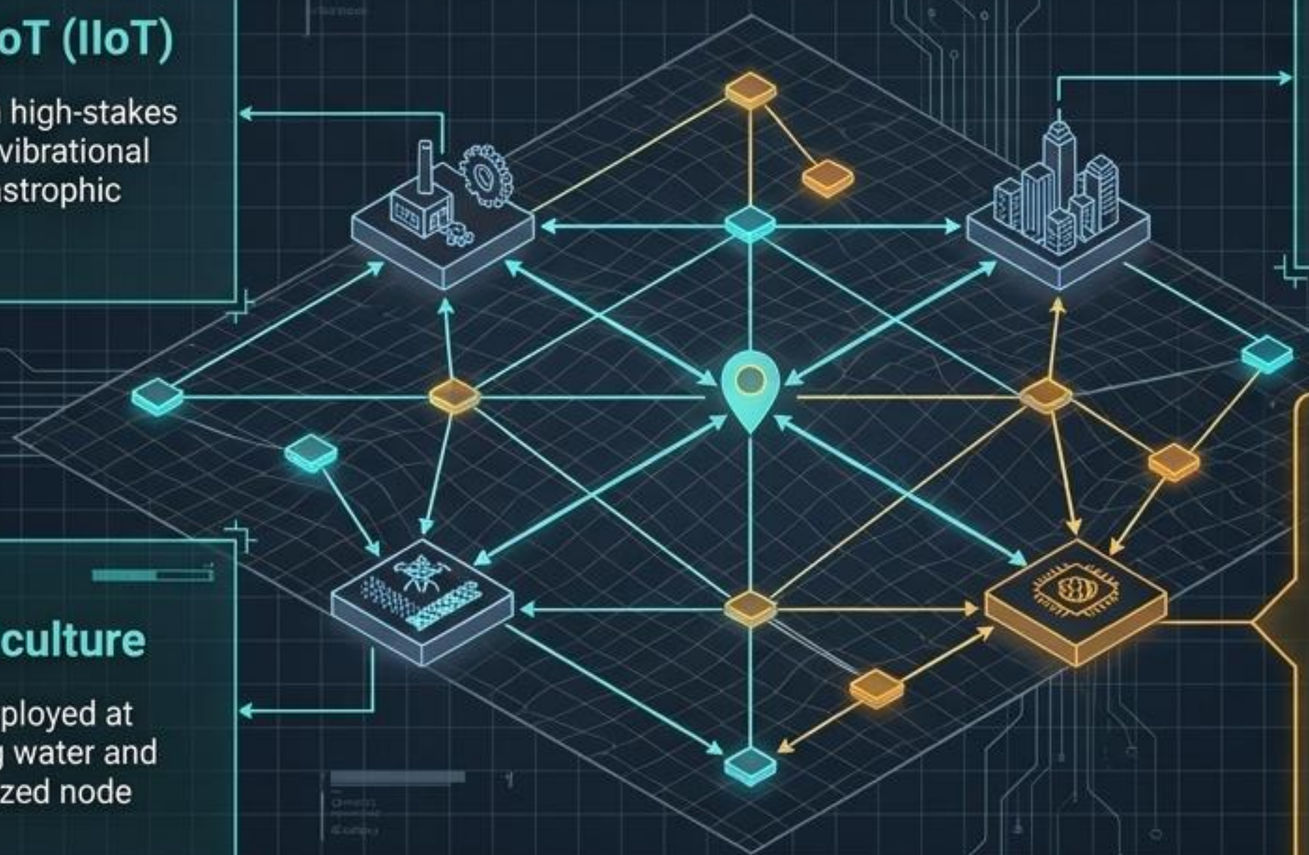
The Global Mesh: Applications Shaping the Autonomous Future

Hub 1:
Industrial IoT (IIoT)

Predictive maintenance in high-stakes factories. Sensors detect vibrational anomalies to prevent catastrophic machinery failures.

Hub 3:
Smart Cities

Intelligent power grids and adaptive traffic management responding in real-time to population flow.

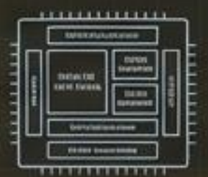


Hub 2:
Smart Agriculture

Environmental sensing deployed at massive scale. Optimizing water and soil pH through decentralized node networks.

Wearable Edge AI

Utilzing specialized silicon variants to Moving TinyML directly to the edge for instant decision-making without cloud latency.



ESP32-S3 Branch:
Specialized Neural Network & Vision Processing

You aren't just learning to write code. You are learning to build the nervous system of the modern world.

The 4-Tier Data Journey: Bridging Physical Constraints and Cloud Scale

Zone 1: Physical Environment (Sensors & Actuators)



Gathering raw analog data and interacting with the physical world.

Zone 2: The Edge (Local Intelligence)



Dual-core stability executing 32-bit local logic without network dependency.

Zone 3: The Network Gateway (Low-Power Connectivity)



Managing local routing, MQTT protocols, and rapid Wi-Fi/Bluetooth coexistence.

Zone 4: Cloud & Platforms (Global Analytics)



The final destination for dashboards, massive datasets, and critical alerts.

High-Impact Embedded Systems: Mastering the Edge Constraints

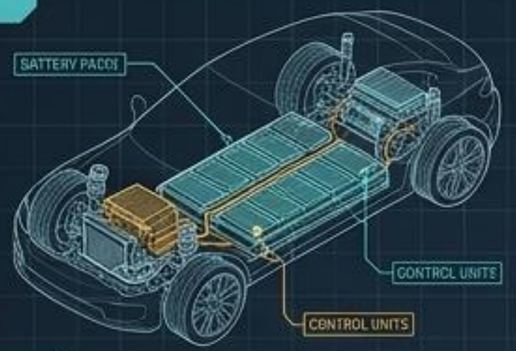
Processing Demands

Power Restraints

Automotive/EV Telemetry

Focus: Real-time motor control.

Constraint: Unforgiving physical environments and real-time electrical monitoring.



Autonomous Robotics



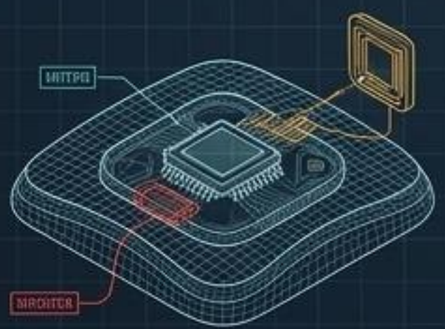
Focus: Zero-latency processing.

Constraint: Requires dual-core isolation to run complex sensor logic without dropping network packets.

Medical Implants

Focus: Life-critical reliability.

Constraint: Micro-amp survival. Dropping draw from 260mA to ~10µA.



Smart Consumer Electronics

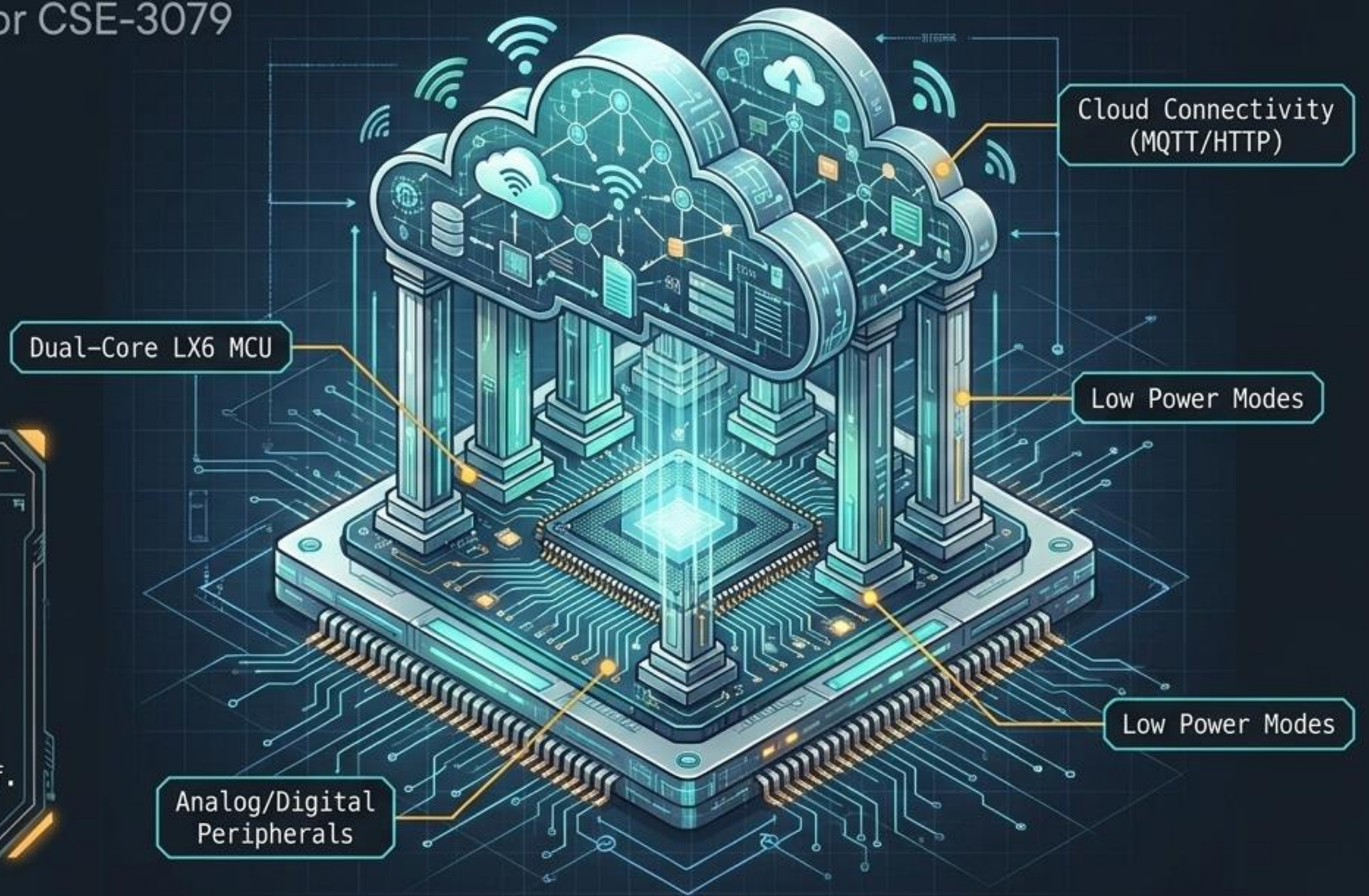
Focus: High-capacity multitasking.

Constraint: Managing complex memory hierarchies and responsive UIs.



Course Orientation: Building the Internet of Things from the Silicon Up

The Fall 2026 Roadmap for CSE-3079

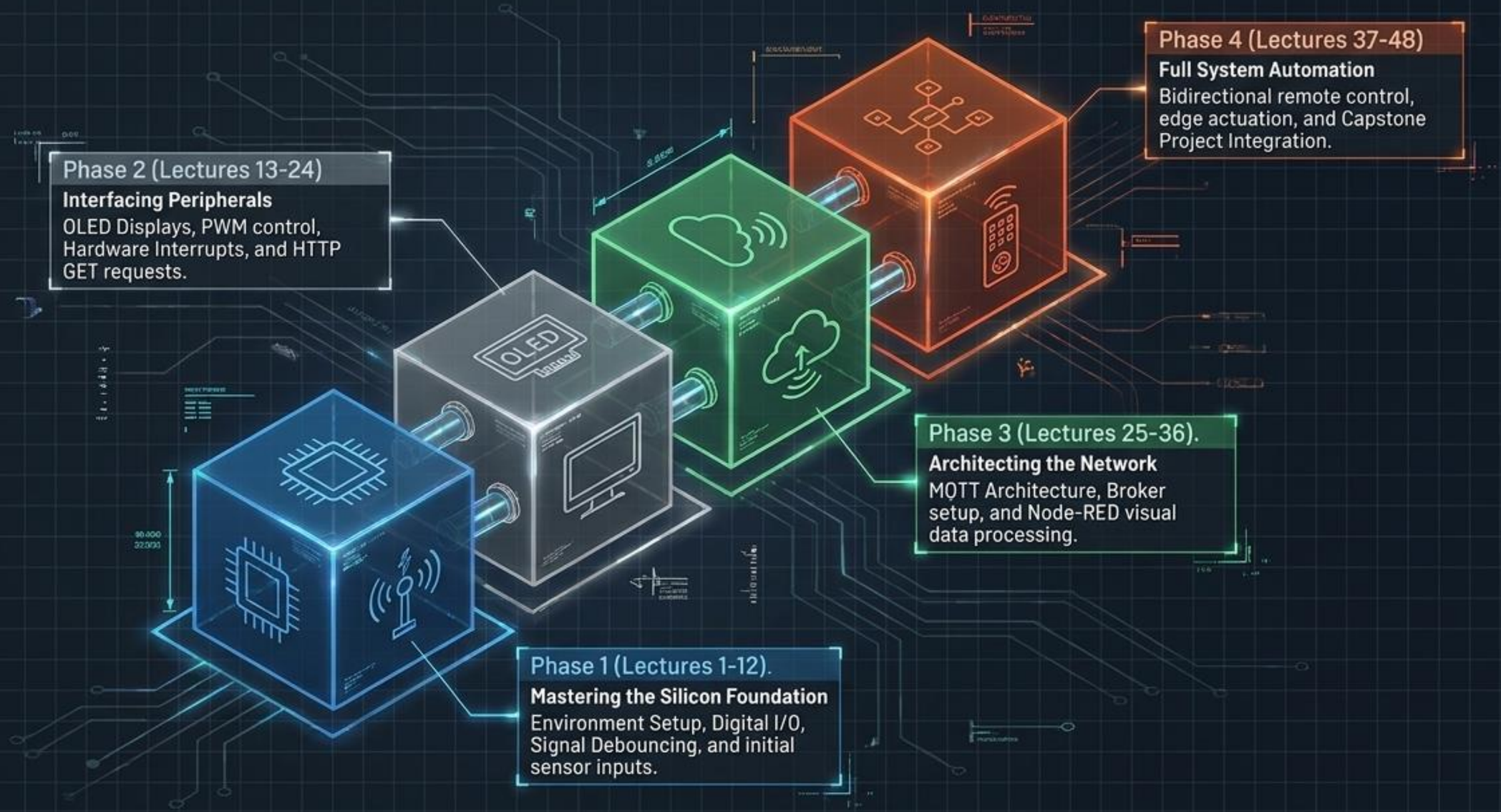


Spec Box

Curriculum Strategy:
48 Lectures.
4 Phases.
1 Complete Edge-to-Cloud System.

We will not use pre-built black boxes. You will construct every layer of the architecture yourself.

The 4-Phase Developer's Pipeline



Phase 4 (Lectures 37-48)

Full System Automation

Bidirectional remote control, edge actuation, and Capstone Project Integration.

Phase 2 (Lectures 13-24)

Interfacing Peripherals

OLED Displays, PWM control, Hardware Interrupts, and HTTP GET requests.

Phase 3 (Lectures 25-36).

Architecting the Network

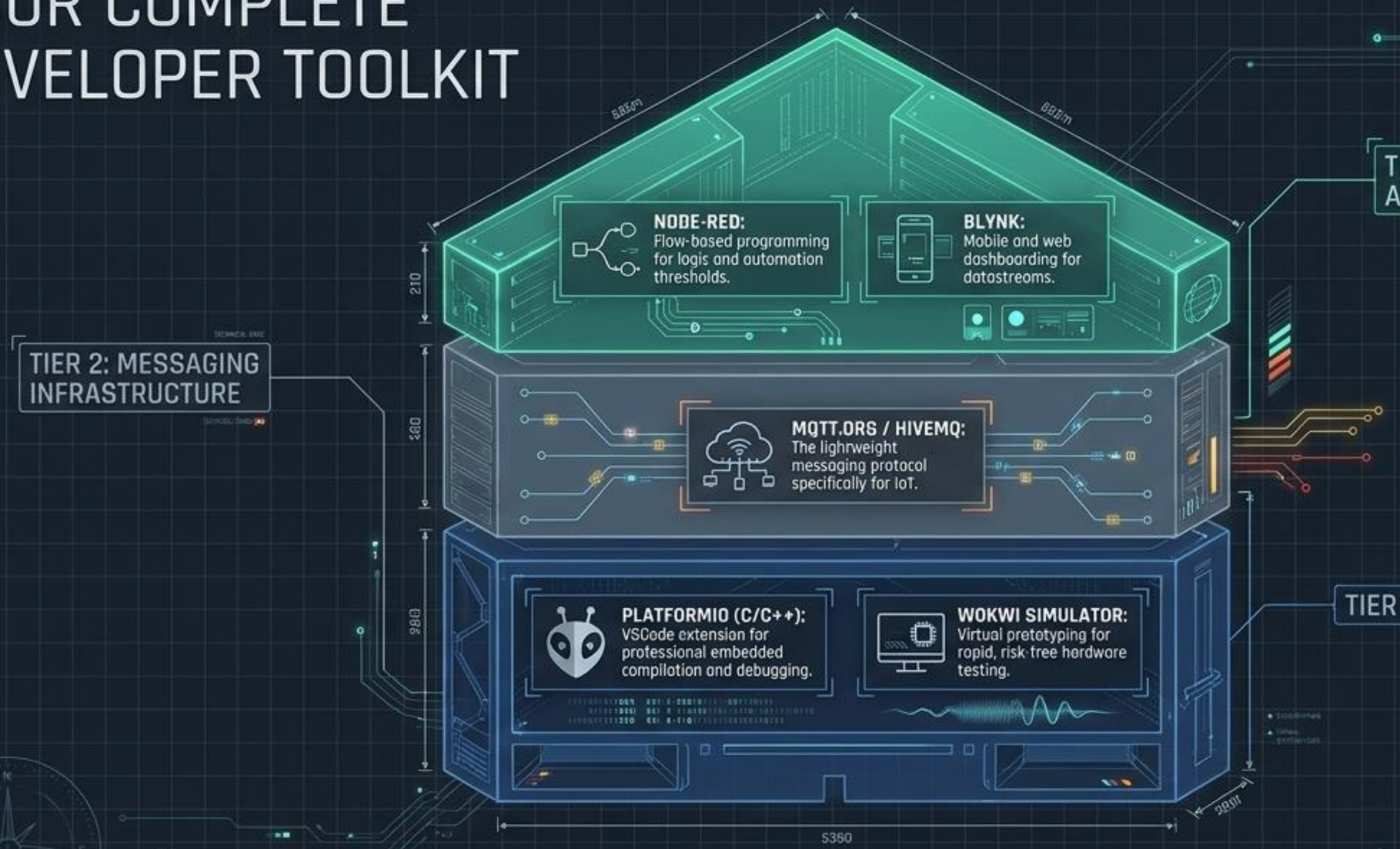
MQTT Architecture, Broker setup, and Node-RED visual data processing.

Phase 1 (Lectures 1-12).

Mastering the Silicon Foundation

Environment Setup, Digital I/O, Signal Debouncing, and initial sensor inputs.

YOUR COMPLETE DEVELOPER TOOLKIT



EVERY TOOL SELECTED FOR DIRECT RELEVANCE TO MODERN INDUSTRIAL ENGINEERING.

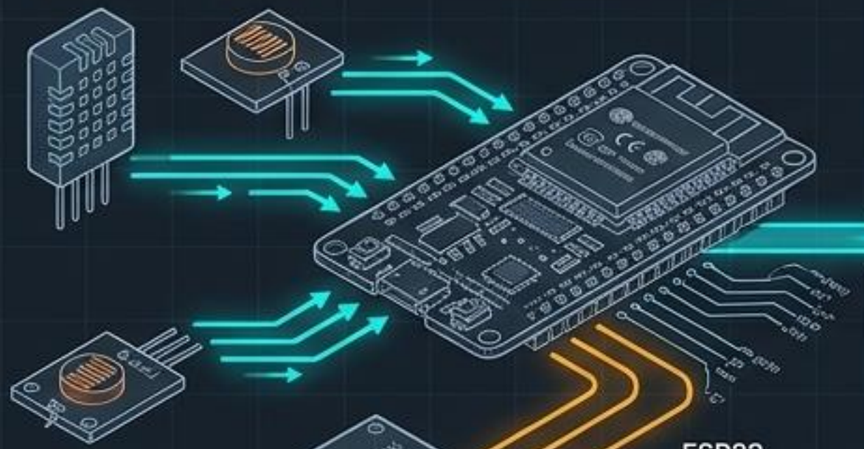
THE BLUEPRINT REALIZED: END-TO-END SYSTEM ARCHITECTURE

PHYSICAL ENVIRONMENT (SENSORS & ACTUATORS)

DHT11 SENSOR MODULE

LDR SENSOR

ESP32 MICROCONTROLLER UNIT



LDR SENSOR

ESP32 MICROCONTROLLER UNIT

PHYSICAL RELAY ACTUATOR

LOCAL GATEWAY & ROUTING

WI-FI 2.4GHz TRANSMISSION



WI-FI 2.4GHz TRANSMISSION

MQTT BROKER SERVER (mosquitto)



MQTT BROKER SERVER (mosquitto)

THE CLOUD (PROCESSING & UI)

NODE-RED LOGIC FLOWCHART



NODE-RED LOGIC FLOWCHART

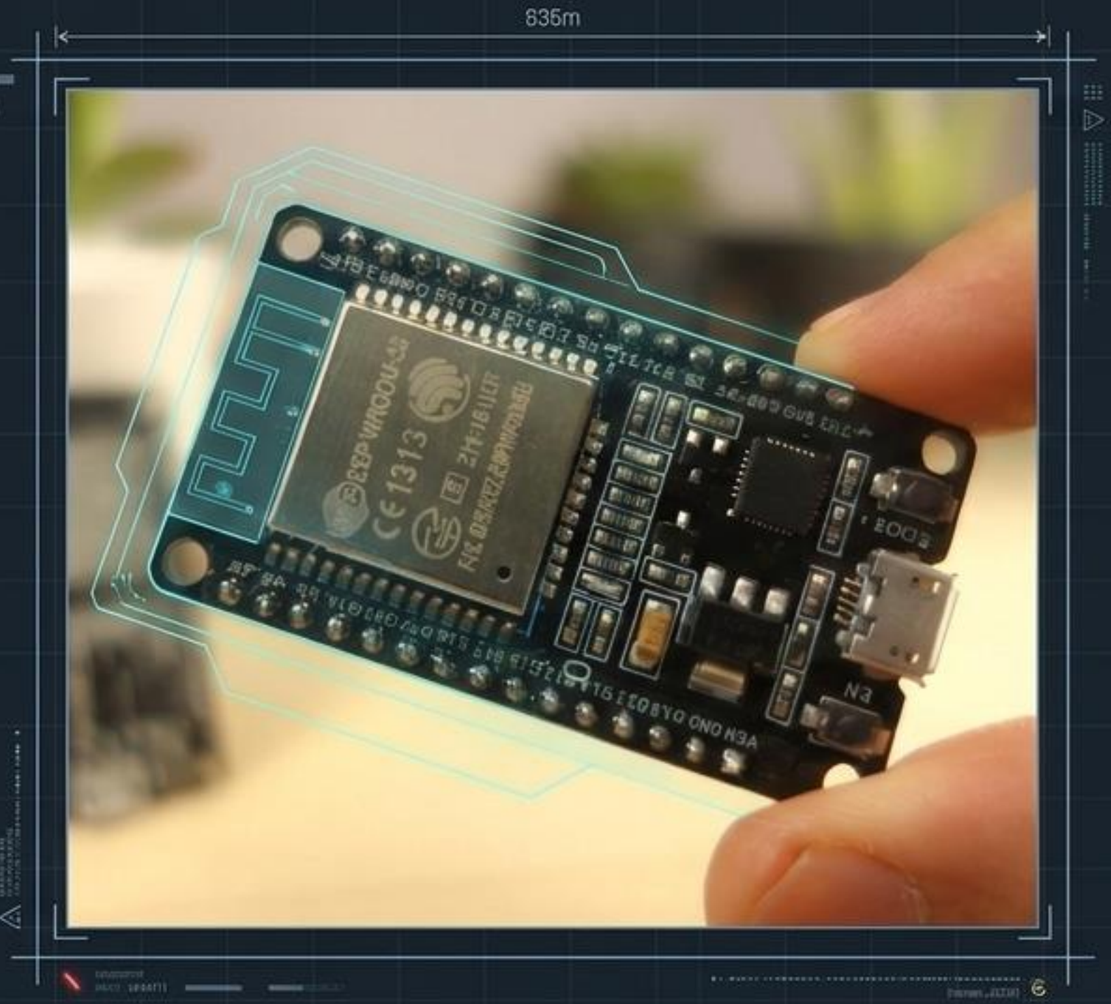
BLYNK SMARTPHONE DASHBOARD



BIDIRECTIONAL CONTROL SIGNAL & ACKN

YOU ARE READY TO BUILD THE FUTURE.
LET'S LOOK AT THE ENGINE.

Lecture 01: The Core Engine Driving Your Architecture

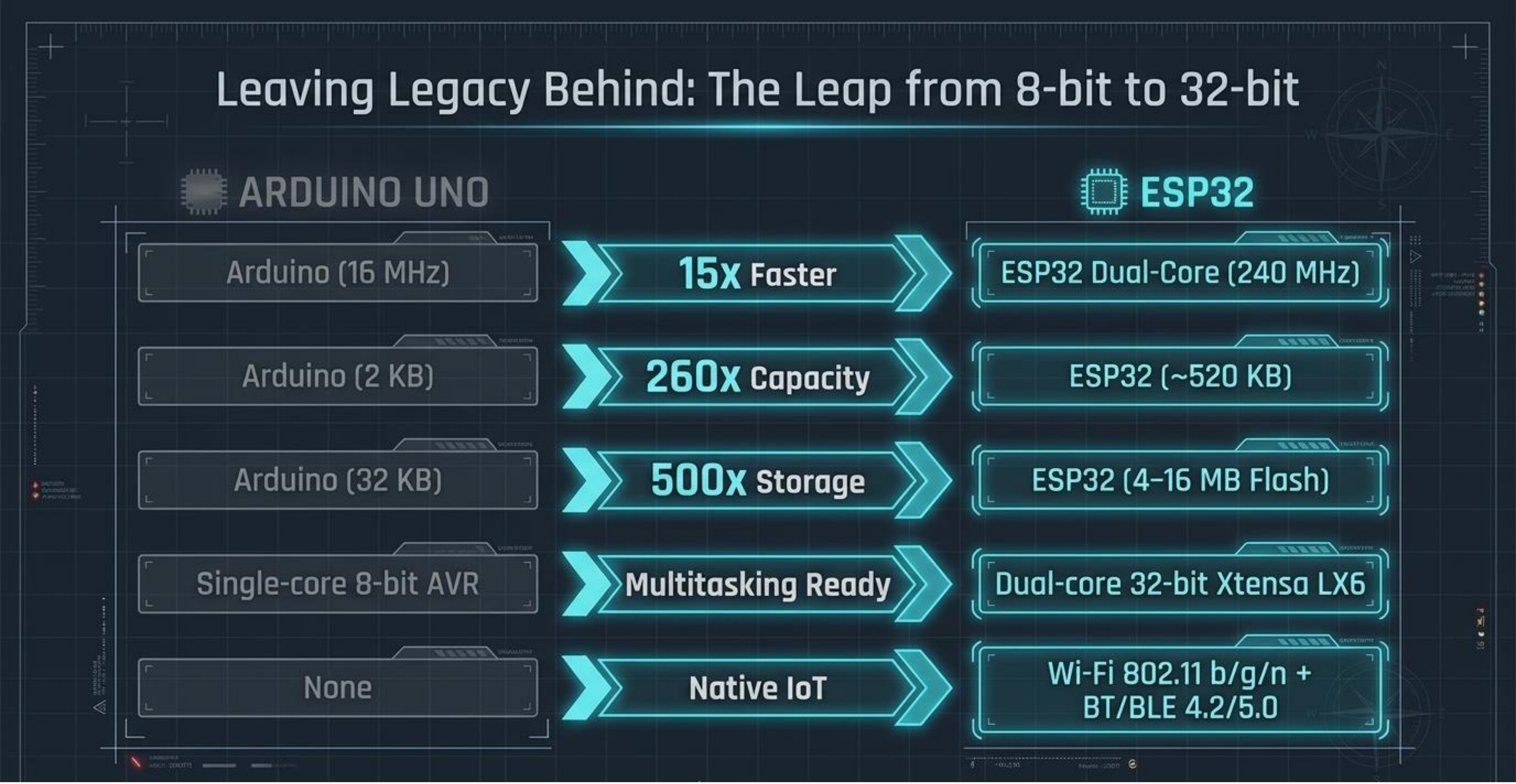


SPEC

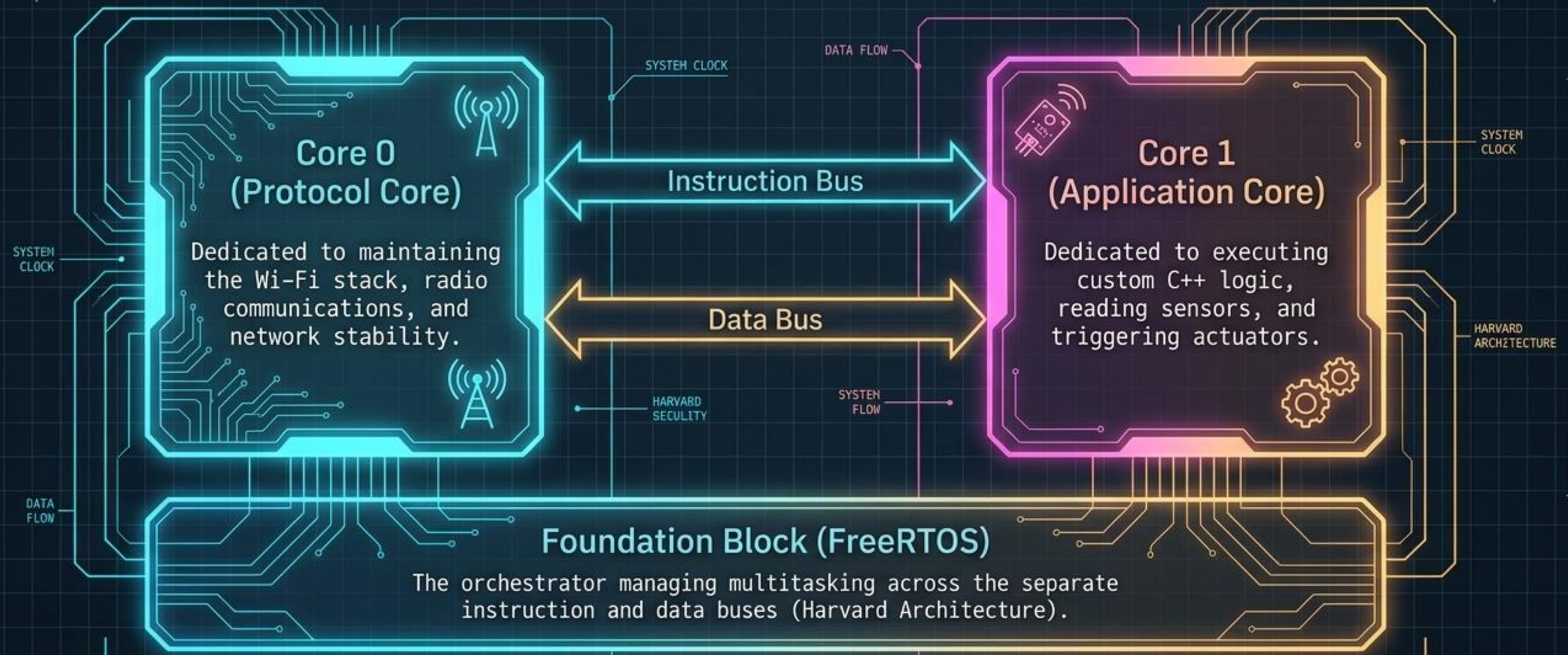
The ESP32 is an ultra-low-power, dual-core powerhouse designed specifically for edge computing. Today, we map its silicon.

> Initiating Silicon Deep-Dive Sequence...

Leaving Legacy Behind: The Leap from 8-bit to 32-bit



Dual-Core Stability: Never Drop a Packet



The Memory Funnel: Speed vs. Capacity

External Flash / PSRAM (4–16 MB)

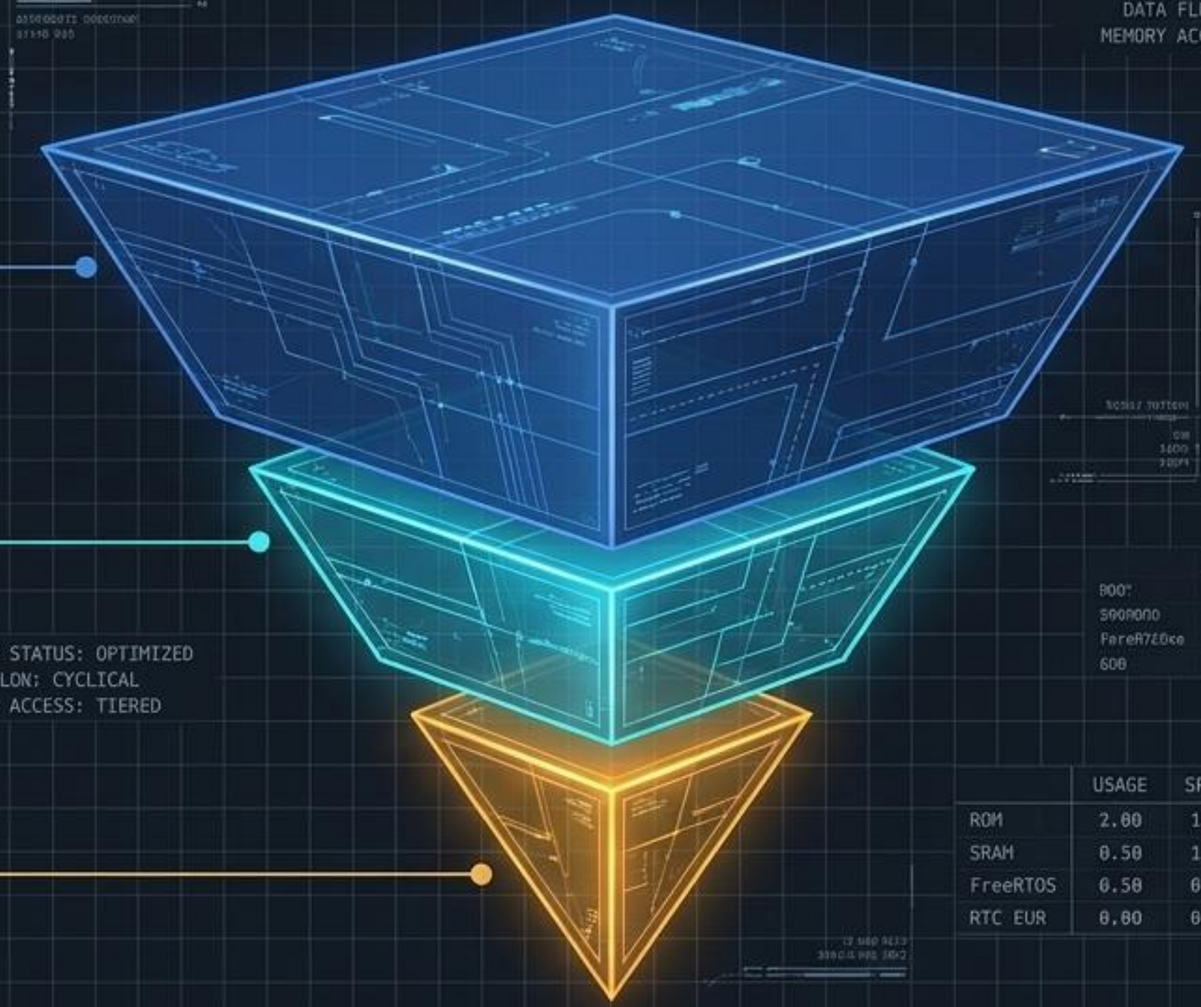
Massive capacity. Non-volatile storage for compiled program code, web server files, and OTA updates.

SRAM (~520 KB)

Fast, volatile runtime memory for variables, buffers, and FreeRTOS active execution.

ROM & RTC Memory

Hyper-fast, permanent. Unalterable bootloader and fundamental Wi-Fi stack. RTC stays alive during deep sleep.

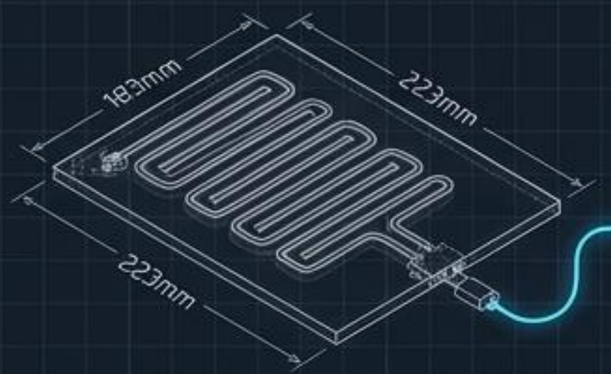


SYSTEM STATUS: OPTIMIZED

SYSTEM W: CMTIMIZED

DATA FLOW: CYCLICAL

RADIO COEXISTENCE ON THE 2.4 GHZ BAND



SERPENTINE PCB
ANTENNA MODULE.



THE PROBLEM

Wi-Fi (802.11 b/g/n) and Bluetooth (Classic + BLE) both operate on the crowded 2.4 GHz frequency band.

FREQ: 2.4 GHz | CHANNEL: CROWDED | STATUS: INTERFERENCE RISK

THE MECHANISM

Time-Division Multiplexing.

The ESP32 hardware rapidly switches the single antenna between the Wi-Fi MAC and the Bluetooth MAC.

SWITCHING FREQ: HIGH | MODE: TDM | STATUS: ACTIVE MULTIPLEXING

THE RESULT

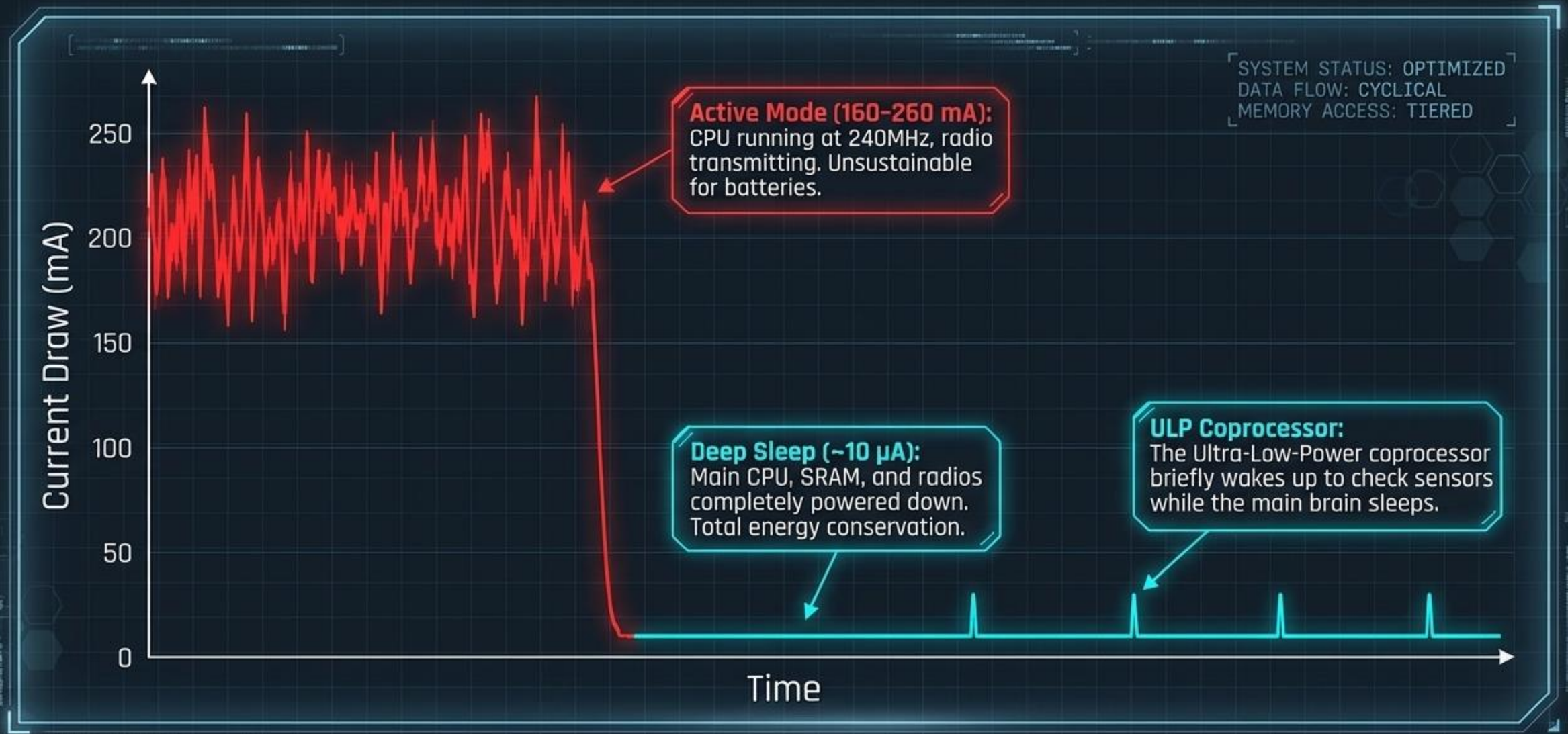
Appears simultaneous.

Allows the ESP32 to act as a BLE beacon scanner while actively streaming data over a Wi-Fi HTTP/MQTT connection.

THROUGHPUT: OPTIMIZED | LATENCY: MINIMAL | STATUS: COEXISTENCE ENABLED

GEMINI NOTEBOOK

Power Modes Dictate Survival: Dropping from 260mA to 10µA



SYSTEM STATUS: SECURED

The Vault: Hardware-Accelerated Security



Cryptographic Acceleration

Hardware-level AES, SHA, RSA, and ECC processing. Encrypts MQTT payloads without slowing down sensor logic.

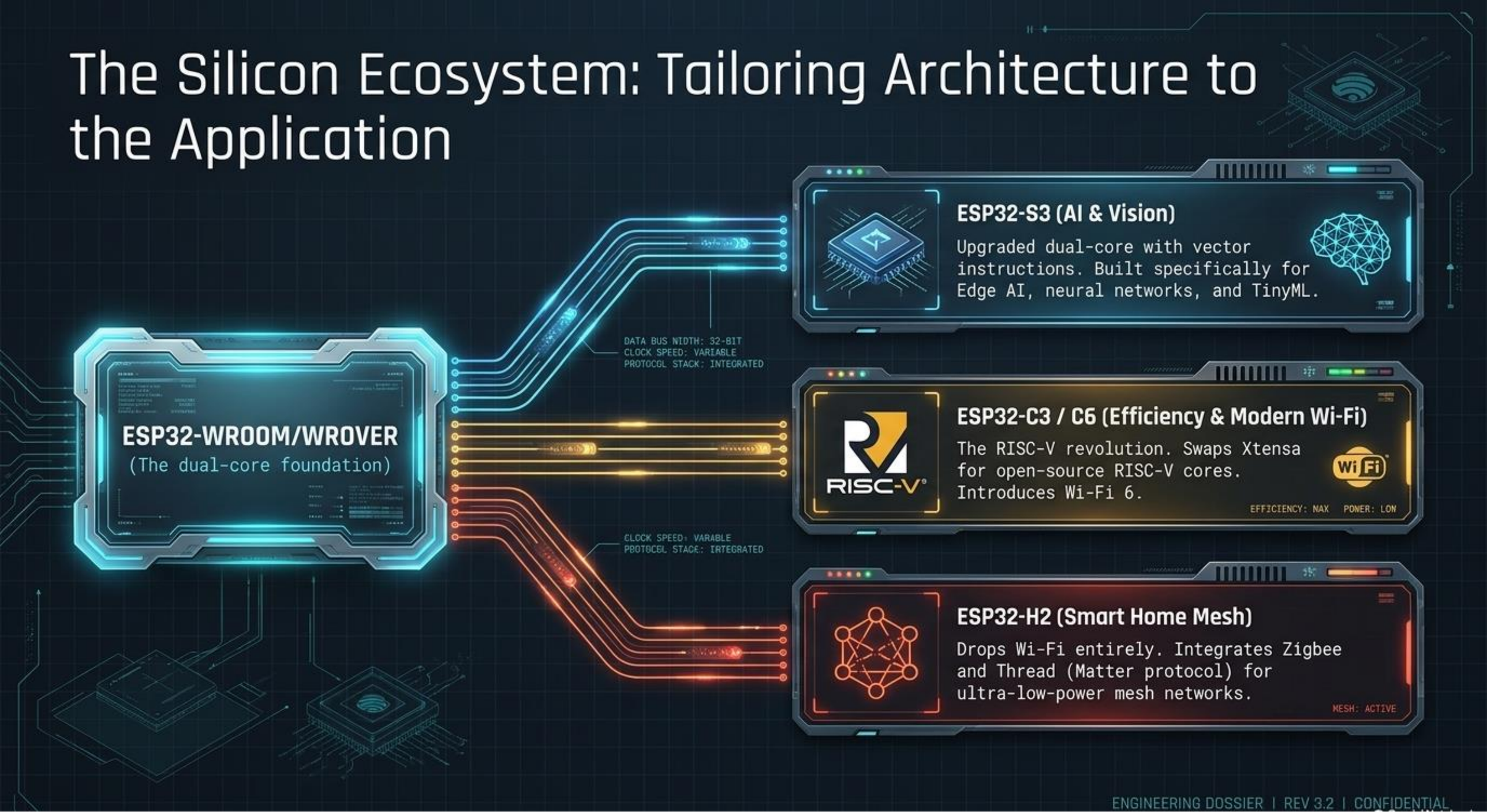
Secure Boot

Ensures the device only runs firmware signed by your trusted cryptographic key. Prevents malicious code injection.

Flash Encryption

Transparently encrypts external Flash memory, protecting proprietary code and hardcoded credentials from physical extraction.

The Silicon Ecosystem: Tailoring Architecture to the Application



ENGINEERING DOSSIER | REV 3.3 | CONFIDENTIAL

Translating Silicon to the Physical Footprint (DevKit V1)

1 The Metal Shield (Faraday Cage)

Houses the module, shielding dual-core CPU and memory from electromagnetic interference.

2 PCB Antenna

Serpentine copper trace handling 2.4GHz Wi-Fi/BLE.



3 Voltage Regulator (AMS1117)

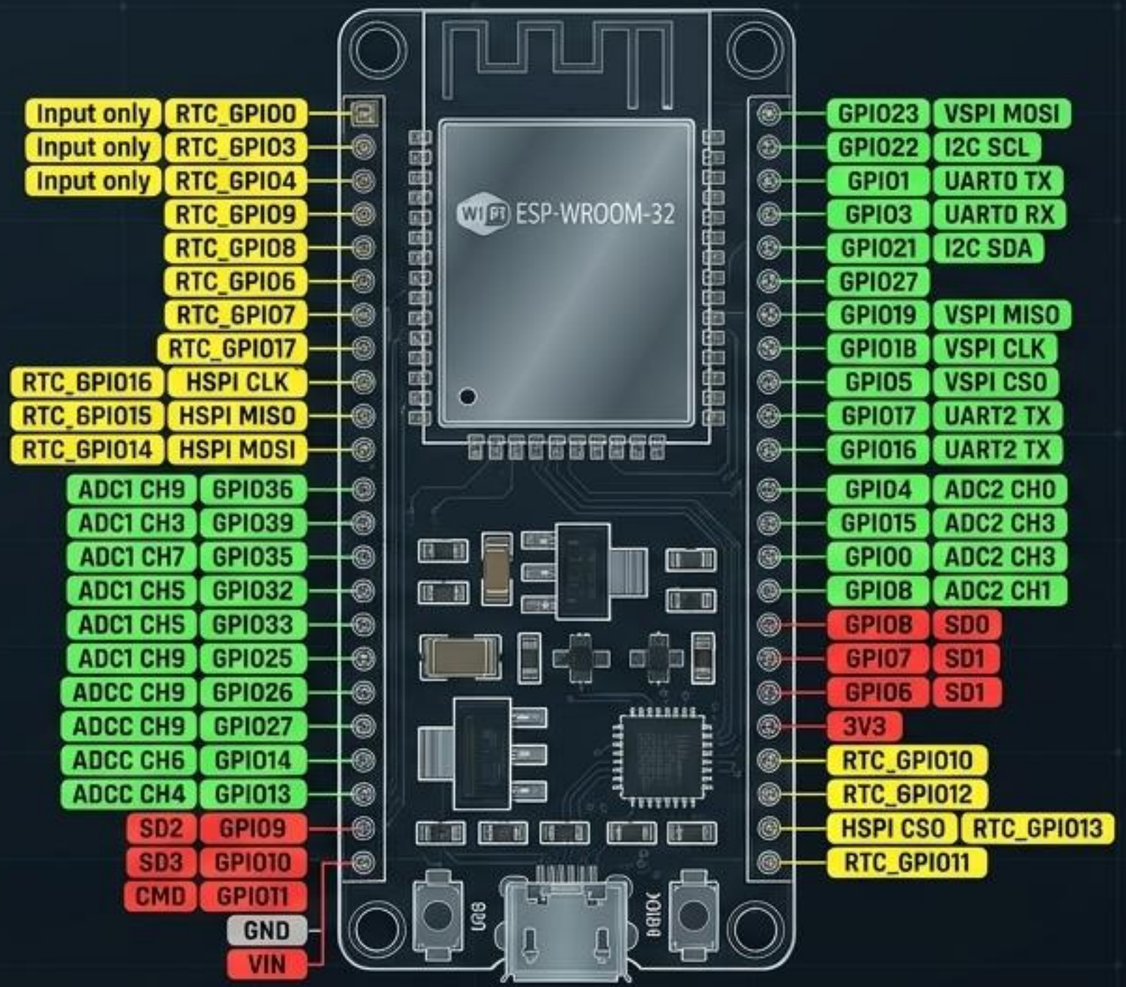
3.3V

Steps down 5V USB input to strict 3.3V operating voltage.

4 CP2102/CH340 Chip

The USB-to-UART bridge allowing direct flashing from a laptop.

The 36-Pin Rules of Engagement: Not all GPIOs are equal



GREEN (Safe to Use)

Standard GPIOs with internal pull-ups/pull-downs. Safe for general input/output, ADC, PWM, I2C/SPI (e.g., 4, 13, 21, 22).

YELLOW (Caution Required)

Special behavior pins. Includes Strapping Pins (0, 2, 5, 12, 15) which dictate boot behavior, and Input-Only pins.

RED (Do Not Touch)

Connected to integrated SPI flash. Touching these will crash the microcontroller instantly.

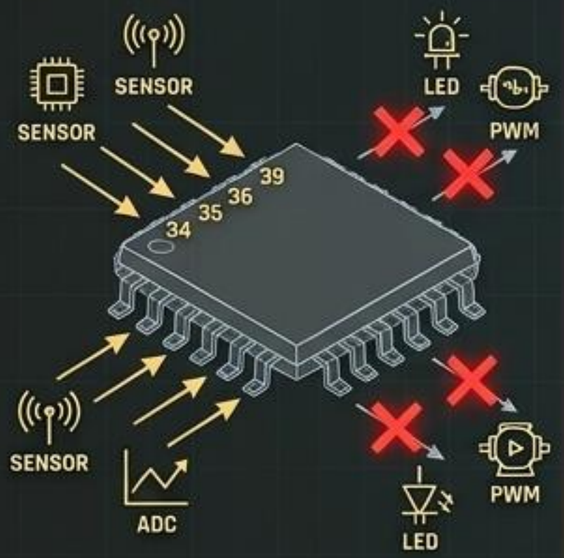
Critical Constraints: Mastering the Hazard Zones

⚠ Constraint 1: The Input-Only Block

Target:
GPIOs 34, 35, 36, 39.

Rule:
Can only read signals (Sensors). Cannot output (No LEDs, no PWM).

Hardware Reason:
They physically lack internal pull-up/pull-down resistors and output driver circuitry.

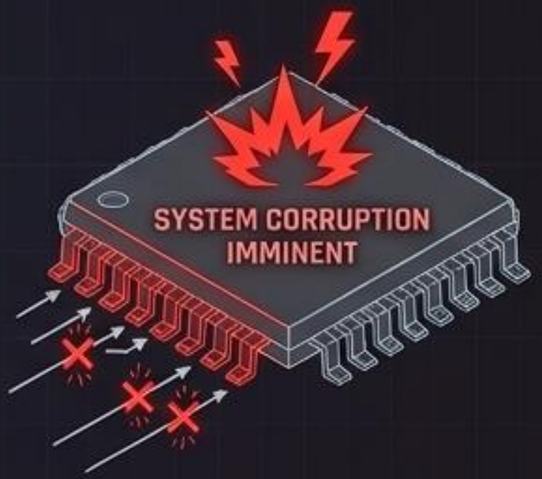


⚠ Constraint 2: The Forbidden Zone

Target:
GPIOs 6 through 11.

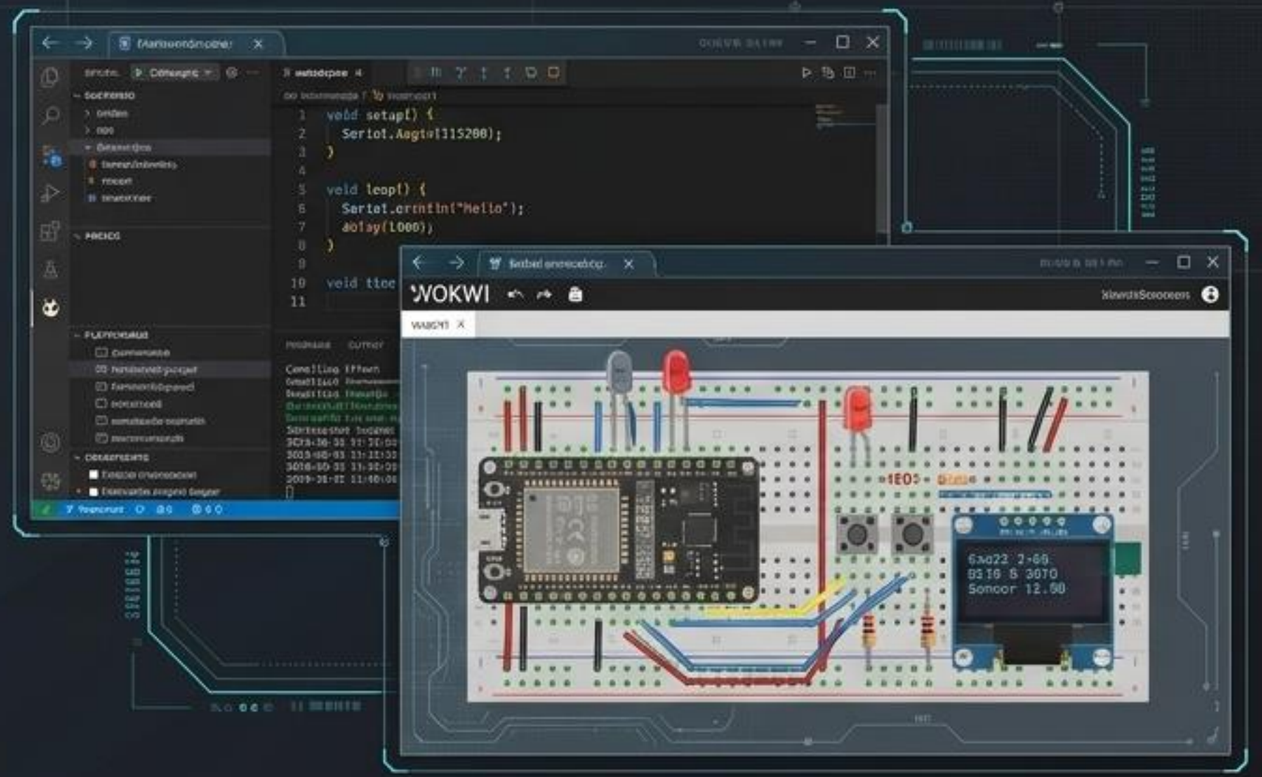
Rule:
NEVER connect external components.

Hardware Reason:
Physically hardwired inside the metal casing to the SPI Flash memory. Interfering causes imminent system corruption.



Initialization Sequence for Lecture 02

- [x] Review Toolchain Documentation (PlatformIO & Wokwi).
- [x] Environment Setup (Install Visual Studio Code on local machine).
- [x] Next Objective: PlatformIO Project Setup, ESP32 Configuration, and first virtual prototyping with Wokwi in Lecture 02.



"Mastery of the edge begins with mastery of the silicon. Understand the architecture, and the code will follow."